ERCOT Ancillary Services (Responsive Reserve Service) Price Forecast

Github Repository:
https://github.com/judyhuazhu/Ju_Wu_Zhu_ENV797_TSA_FinalProject

Skyler Ju, Joy Wu, Judy Hua Zhu

Introduction, Motivation, Relevance, Objectives

Write a few paragraphs detailing the rationale for your study. This should include both the context of the topic as well as a rationale for your choice of dataset (reason for location, variables, etc.). You may choose to include citations if you like or any other reference you may have used during the project (optional).

The Electric Reliability Council of Texas (ERCOT) manages the power grid for roughly 90% of the state's electric load. Unlike most U.S. wholesale electricity markets, ERCOT operates an energy-only market with no capacity payments, meaning that generators and flexible resources like battery storage earn revenue primarily through energy sales and ancillary services. Ancillary services (AS) are products procured by ERCOT to maintain grid reliability in real time, covering functions such as frequency regulation, spinning reserves, and contingency response. ERCOT currently procures five types of AS in its Day-Ahead Market: Regulation Up (REGUP), Regulation Down (REGDN), Responsive Reserve Service (RRS), ERCOT Contingency Reserve Service (ECRS), and Non-Spinning Reserve (NSPIN). Each type serves a different operational need, from second-by-second frequency corrections to 30-minute backup capacity for sustained capacity.

Like we mentioned above, AS has became the primary revenue stream in ERCOT for the energy-only market. In fact, AS made up 87% of battery storage revenues in ERCOT in early 2023. The growing renewable deployment, including solar and wind, creates more variability issues. This leads to higher AS demand in the market because they help to reduce operational risks, address intermittency issues, and maintain grid reliability.

Therefore, being able to forecast AS prices matter in this context, as storage operators need forecasts to decide when to bid into AS or energy markets. AS prices are also highly volatile, and developers need AS revenue projections for investment decisions.

This project focuses specifically on RRS (Responsive Reserve Service) because RRS prices are highly volatile, with hourly clearing prices spiking above $1,000/MW during scarcity events such as generator trips or extreme heat. It is directly tied to grid reliability during large disturbances, since it's mainly used for these events.

The objective of this analysis is to develop and evaluate time series forecasting models for RRS Market Clearing Price for Capacity (MCPC) in the ERCOT Day-Ahead Market. Using hourly price data from December 2023 through April 2026, this study trains models on the December 2023 through December 2025 period and tests forecast accuracy against January through April 2026. The analysis compares various models, and assesses whether the structural price decline observed in 2024 and 2025 creates particular challenges for time series forecasting methods.

Dataset information

Provide information on how the dataset for this analysis were collected (source), the data contained in the dataset (format). Describe how you wrangled/processed your dataset to get the time series object.

Add a table that summarizes your data structure (variables, units, ranges and/or central tendencies, data source if multiple are used, etc.). This table should inserted as a `kable` function in an R chunk. Just show the first 10 rows of your data. Do not include the code used to generate your table.

The dataset used in this analysis was obtained from the ERCOT public website under its market information –> Day-Ahead Market AS prices: https://www.ercot.com/mp/data-products/data-product-details?id=NP4-188-CD. The raw data contains hourly Market Clearing Prices for Capacity (MCPC) in $/MW for all five ancillary service types traded in ERCOT's Day-Ahead Market. The dataset spans from December 13, 2023 through April 8, 2026, covering 848 calendar days.

The raw data was in long format with one row per delivery date, hour ending, and ancillary service type combination. Each row contained the delivery date, hour ending (01:00 through 24:00 in ERCOT convention), ancillary service type (ECRS, NSPIN, REGDN, REGUP, or RRS), the MCPC value, and a DST flag column.

We took several wrangling steps to prepare the data for time series analysis. First, Daylight Saving Time entries were handled. Fall-back days in November had duplicate 02:00 hour entries (one flagged as DST) which were averaged to produce a single value per hour. The data was then pivoted from long to wide format using `pivot_wider()` so that each row represents a unique date-hour combination with the five ancillary service types as separate columns. After pivoting, spring-forward days in March were missing the 02:00 hour entirely. A complete hourly grid was generated using `expand.grid()` and joined to the pivoted data, and the missing hours were filled using linear interpolation via `na.approx()`. The final processed dataset contains 20,352 hourly observations (848 days times 24 hours) with seven columns: Date, hourEnding, and five MCPC columns.

The dataset was split into a training period (December 13, 2023 through December 31, 2025) and a test period (January 1, 2026 through April 8, 2026). All model fitting, decomposition, and diagnostic analysis were performed on the training set only.

Table 1: First 10 Rows of Processed ERCOT Ancillary Services Dataset

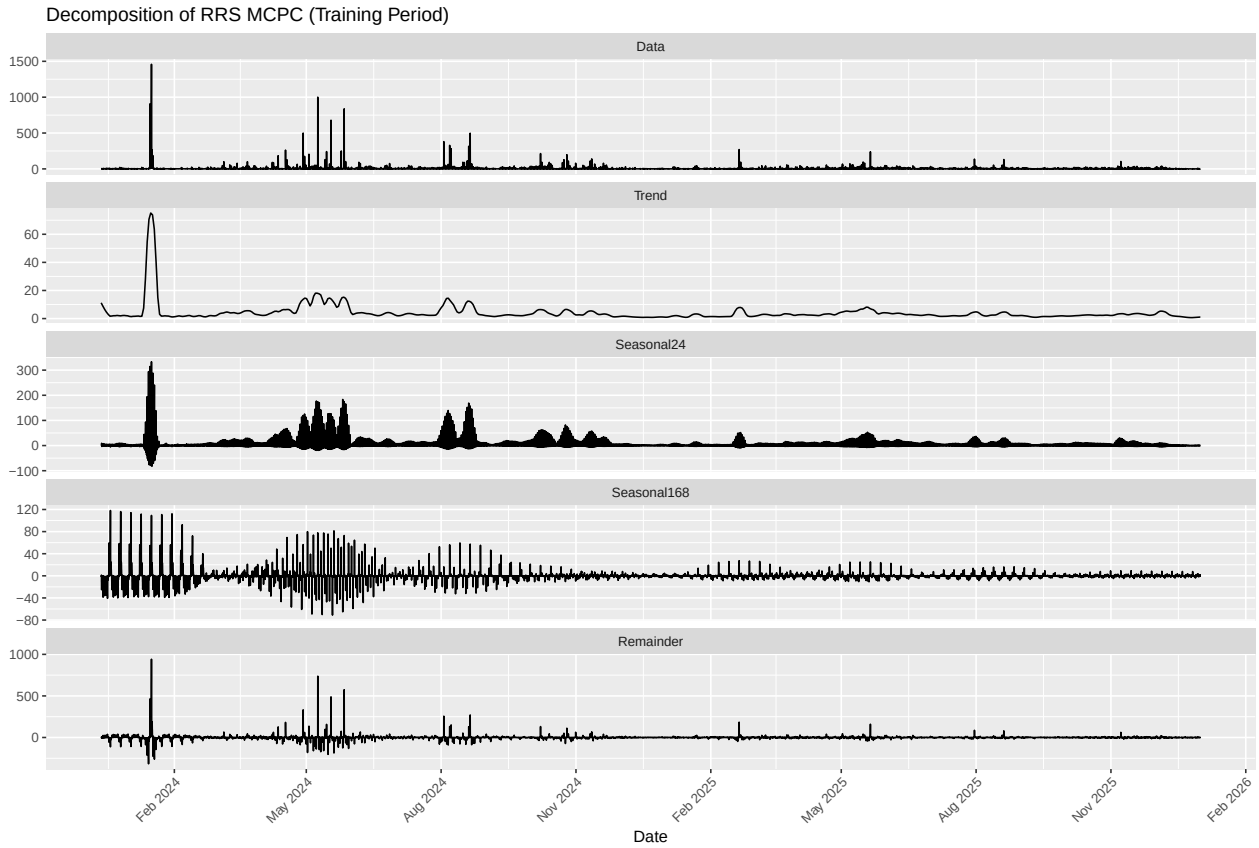
Date	Hour Ending	ECRS ($/MW)	NSPIN ($/MW)	REGDN ($/MW)	REGUP ($/MW)	RRS ($/MW)
2023-12-13	01:00	1.57	1.03	1.51	1.90	1.57
2023-12-13	02:00	1.51	1.04	2.00	1.97	1.51
2023-12-13	03:00	1.50	1.01	1.25	1.94	1.50
2023-12-13	04:00	1.58	1.04	1.25	2.00	1.58
2023-12-13	05:00	1.58	0.99	1.25	2.51	1.71
2023-12-13	06:00	0.10	1.17	1.56	4.46	2.00
2023-12-13	07:00	2.00	3.00	2.00	4.99	2.00
2023-12-13	08:00	3.09	3.00	2.00	3.98	2.09
2023-12-13	09:00	2.00	3.00	6.94	3.48	2.00
2023-12-13	10:00	7.50	2.72	5.25	4.50	4.75

Table 2: Variable Summary: ERCOT Day-Ahead AS MCPC, Training Period (Dec 2023–Dec 2025)

Variable	Description	Unit	Observations	Mean	Median	SD	Min	Max
ECRS	ERCOT Contingency Reserve Service	$/MW	18,000	5.28	1.00	39.14	0.02	2,086.26
NSPIN	Non-Spinning Reserve	$/MW	18,000	4.58	1.01	34.01	0.04	2,150.00
REGDN	Regulation Down	$/MW	18,000	2.52	1.35	16.27	0.00	1,455.29
REGUP	Regulation Up	$/MW	18,000	4.45	1.50	25.42	0.00	1,007.00
RRS	Responsive Reserve Service	$/MW	18,000	4.16	1.00	29.19	0.17	1,458.75

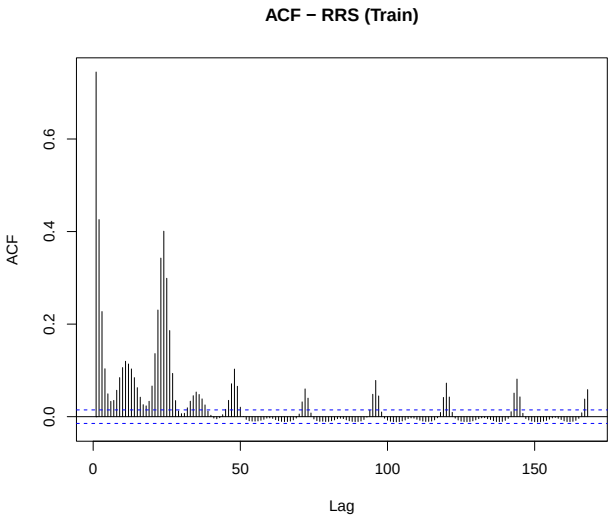
Source: ERCOT Public Website.

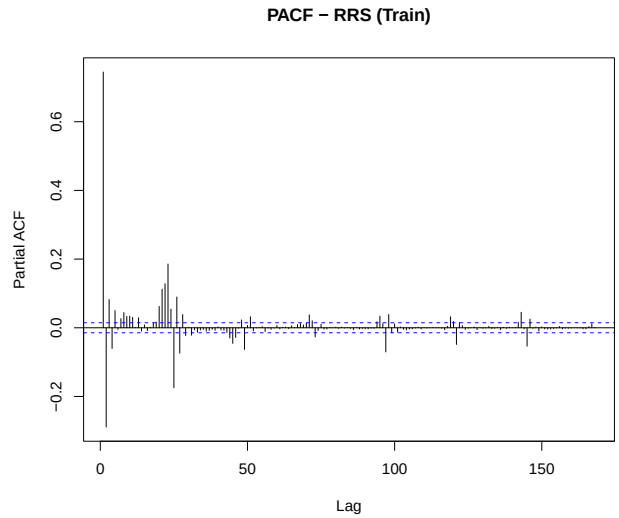
Decomposing the Training Time Series



The series is dominated by large price spikes, especially in early 2024 where hourly RRS prices hit almost \$1,500/MW. By 2025, spikes become much less frequent and the overall price level drops in the trend component. The daily seasonal pattern (Seasonal24) is strongest during summer months when AC-driven demand peaks in afternoon and evening hours. The weekly pattern (Seasonal168) shows lower weekend prices, which is expected since commercial demand drops. The remainder is large relative to the other components, meaning extreme scarcity events drive much of the price variation and will be difficult to forecast.

ACF and PACF on Training Period





The ACF decays slowly across all 168 lags, indicating non-stationarity and the need for differencing. Clear spikes appear at lag 24 (daily cycle) and around lag 168 (weekly cycle). The PACF has a strong spike at lag 1 followed by a sharp drop, suggesting an AR(1) component, with smaller significant spikes at lags 24 and 168 confirming both seasonal cycles. This pattern points toward an ARIMA model with Fourier terms at the daily and weekly frequencies as a good modeling approach.

Analysis (Methods and Models)

Methods

We separated the dataset into train and test pairs, distinguished by time before (18000 obs) and after (2352 obs) 2026. Then trained the data with ARIMA model, ARIMA + Fourier, NNETAR, TBATS, and daily seasonal Naive model. Lastly, all trained models are tested by their RMSE to true values in test set for comparison and discussion.

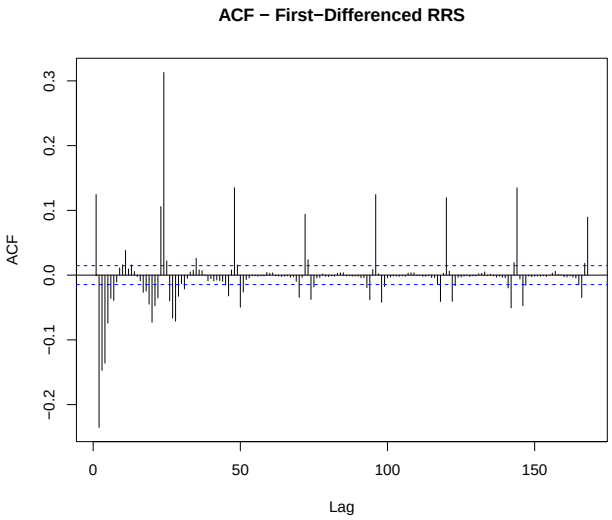
ARIMA

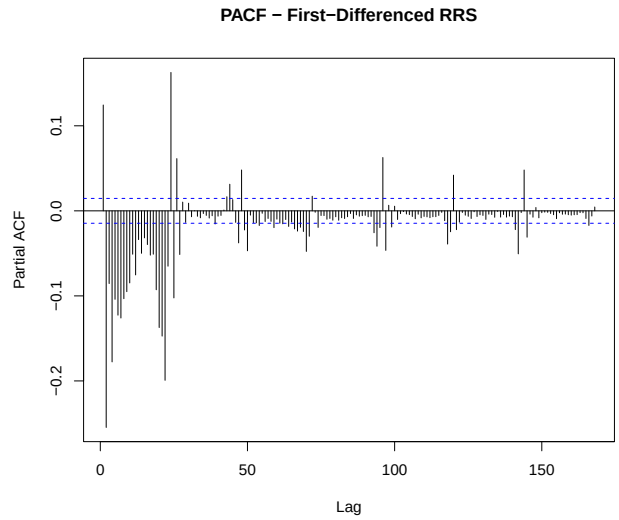
We include three ARIMA-based models: a baseline ARIMA(1,1,1), an ARIMA(1,1,1) with Fourier terms at k=6, and an ARIMA(1,1,1) with Fourier terms at the optimal k selected through grid search. Model order was chosen by inspecting the differenced ACF and PACF series.

Model 1: ARIMA(1,1,1)

Differenced Series: ACF and PACF

The data exploration section showed that the RRS series is non-stationary, so we applied one round of first differencing before inspecting the ACF and PACF to identify the model order.

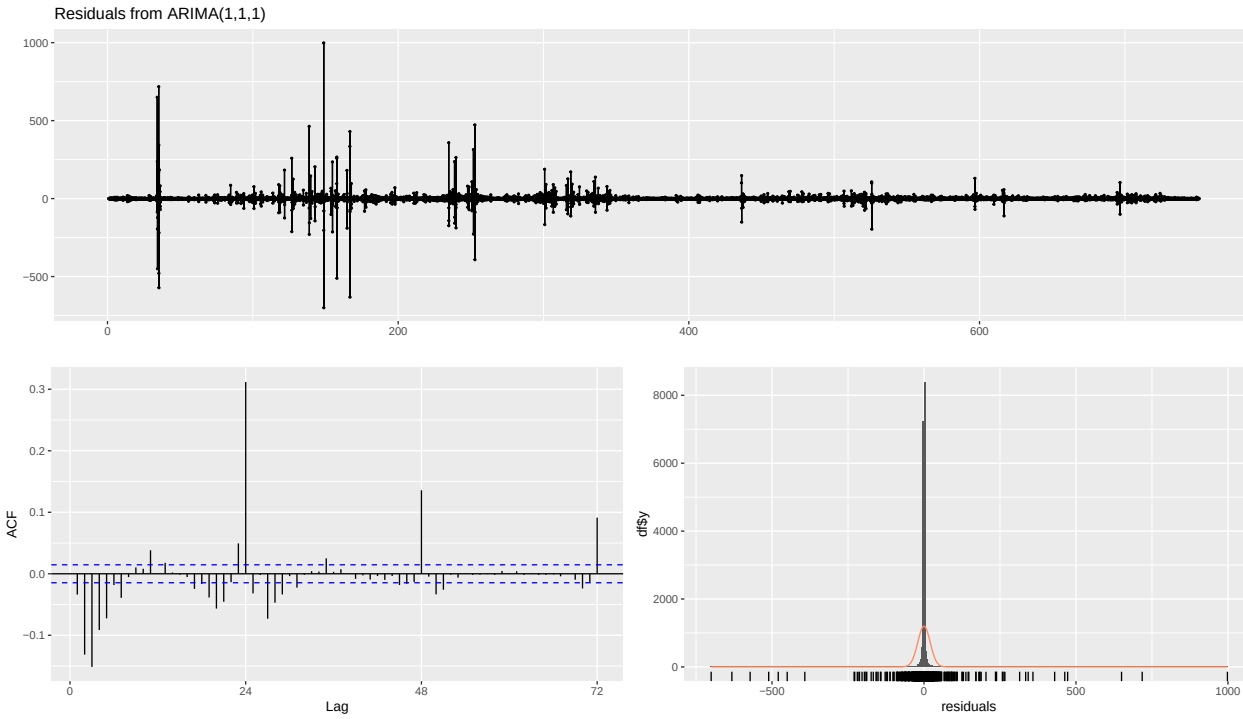




After differencing, the slow ACF decay in the original series is gone, confirming d=1 is sufficient for stationarity. The ACF shows a significant negative spike at lag 1 followed by values mostly within the confidence bounds, consistent with MA(1), so we set q=1. The PACF shows a dominant spike at lag 1 followed by smaller significant spikes at higher lags, so we set p=1. Spikes at lags 24 and 48 are also visible in the ACF, pointing to a residual daily seasonal cycle that first differencing does not remove. It is addressed through Fourier terms in the next step.

Model Fit and Residual Diagnostics

```
## Series: ts_RRS_arima
## ARIMA(1,1,1)
##
## Coefficients:
##           ar1     ma1
##       -0.4480  0.6512
## s.e.   0.0201  0.0168
##
## sigma^2 = 413.8:  log likelihood = -79763.2
## AIC=159532.4   AICc=159532.4   BIC=159555.8
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,1,1)
## Q* = 3546.2, df = 46, p-value < 2.2e-16
##
## Model df: 2.   Total lags used: 48
```

```
##
##  Box-Ljung test
##
## data:  ARIMA_RRS$residuals
## X-squared = 2996.7, df = 24, p-value < 2.2e-16
```

The Ljung-Box test strongly rejects white noise (Q*=3546.2, p<2.2e-16), showing significant autocorrelation remains in the residuals. The residual ACF shows clear spikes at lags 24 and 48, confirming that the daily seasonal cycle is not captured by ARIMA(1,1,1) alone. The histogram is heavily right skewed, reflecting the non-normal distribution driven by extreme price events. These results motivate adding Fourier terms to directly model the daily seasonal cycle in the next step.

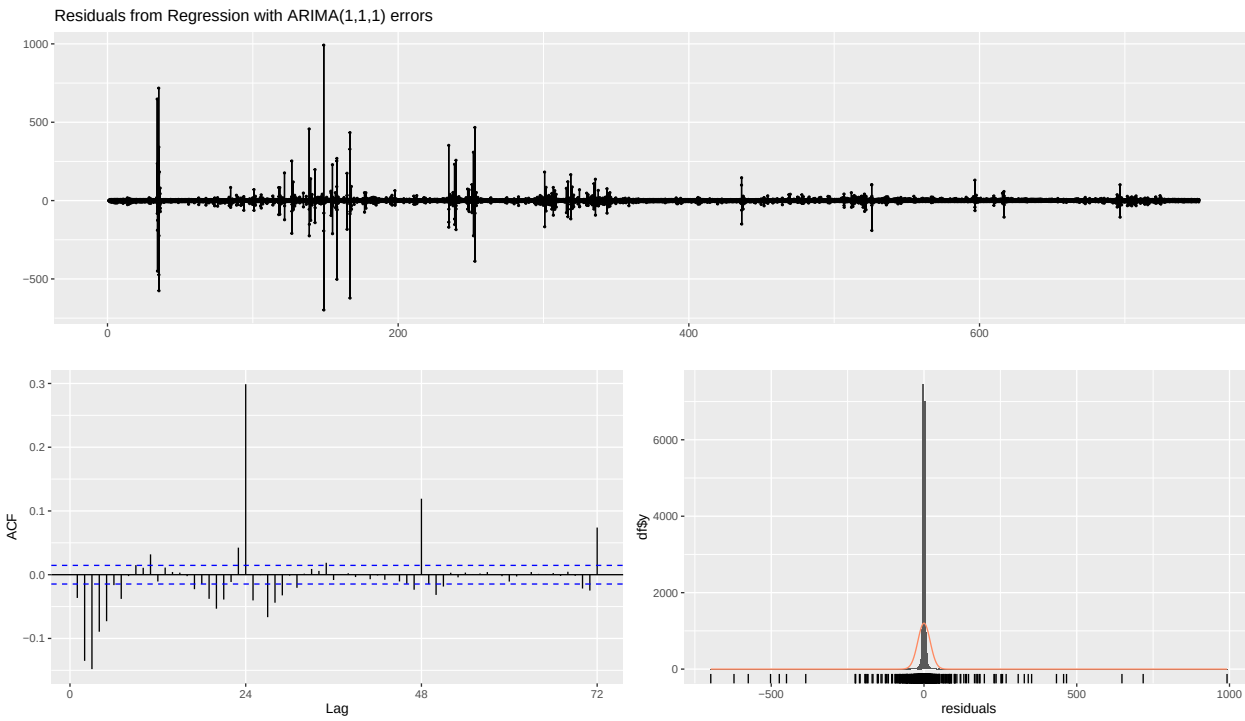
Model 2: ARIMA(1,1,1) + Fourier (k = 6)

The residual diagnostics from ARIMA(1,1,1) show that the daily seasonal cycle at lag 24 is not captured by the base model. We extend the model by adding Fourier terms as external regressors. For a daily cycle of period 24, k ranges from 1 to 12 (period/2). We set k = 6 as a balanced choice between underfitting and overfitting before gridding search all possibilitys between 1<k<12.

Model Fit and Residual Diagnostics

```
## Series: ts_RRS_arima
## Regression with ARIMA(1,1,1) errors
##
## Coefficients:
```

```
##           ar1     ma1    S1-24   C1-24    S2-24    C2-24    S3-24    C3-24
##       -0.4672  0.6606  -2.4830  1.6499  -4.8750  -1.2222  -0.5788  -1.6287
## s.e.   0.0203  0.0170   0.9202  0.9202   0.4629   0.4629   0.3117   0.3117
##        S4-24    C4-24   S5-24    C5-24   S6-24   C6-24
##       1.1865  -2.2655  0.9071  -0.1425  0.8638  0.6713
## s.e.  0.2369   0.2369  0.1925   0.1925  0.1631  0.1631
##
## sigma^2 = 406.3:  log likelihood = -79594.13
## AIC=159218.3   AICc=159218.3   BIC=159335.2
```




```
##
##  Ljung-Box test
##
## data:  Residuals from Regression with ARIMA(1,1,1) errors
## Q* = 3265.2, df = 46, p-value < 2.2e-16
##
## Model df: 2.   Total lags used: 48
```

```
##
##  Box-Ljung test
##
## data:  ARIMA_Fourier_RRS$residuals
## X-squared = 2810.1, df = 24, p-value < 2.2e-16
```

The AIC drops from 159532.4 to 159218.3, showing better overall fit. However, the Ljung Box test still strongly rejects white noise (Q*=2810.1, p<2.2e-16), and spikes at lags 24 and 48 remain in the residual ACF, suggesting the daily seasonal pattern in RRS prices is too irregular to be fully captured by fixed Fourier.

Accuracy on Test Period

```
##                        ME     RMSE      MAE        MPE     MAPE      MASE
## Training set -2.942974e-05 20.33953 2.990471  -20.09339  67.4006 1.0497959
## Test set      1.816705e+00 29.94081 2.671455 -156.16212 180.6153 0.9378063
##                     ACF1
```

```
## Training set -0.03377676
## Test set              NA
```

```
##                         ME     RMSE     MAE         MPE     MAPE     MASE
## Training set -1.034186e-05 20.14937 3.64858   -7.707739 154.7702 1.280823
## Test set     -1.085948e+00 30.06369 4.78752 -487.814187 504.0986 1.680645
##                     ACF1
## Training set -0.03645802
## Test set              NA
```

The Fourier extension marginally improves training set RMSE (20.15 vs. 20.34 for the baseline), but this does not carry over to the test set. Test RMSE increases slightly from 29.94 to 30.06, and test MAPE jumps significantly from 180.6 to 504.1, suggesting K = 6 overfits the training data and does not generalize well to the test period. This motivates the grid search for a better-fitting k.

Model 3: ARIMA(1,1,1) + Fourier (k=2)

Fitting ARIMA + Fourier (grid search most suiatble K)

```
##     K     RMSE      MAE
## 1   1 30.24309 4.027228
## 2   2 30.03612 4.937974
## 3   3 30.04885 4.834251
## 4   4 30.08505 5.466041
## 5   5 30.11243 5.506840
## 6   6 30.06369 4.787520
## 7   7 30.05698 4.423178
## 8   8 30.05383 4.416627
## 9   9 30.06236 4.454361
## 10 10 30.06844 4.545005
## 11 11 30.06122 4.411661
## 12 12 30.06016 4.403254
```

Now K=2 is performing the best, try auto arima since arima parameters may change with K.

```
##                        ME     RMSE      MAE       MPE     MAPE     MASE
## Training set -0.001977189 18.43019 3.441486 -31.79394 169.4512 1.208124
## Test set      2.136921039 30.04493 4.946866 380.98655 856.6338 1.736583
##                       ACF1
## Training set -0.0002023051
## Test set                NA
```

Update the fourier model with best K and ARIMA parameters

```
##                         ME     RMSE      MAE       MPE     MAPE     MASE
## Training set -0.0001621573 20.26900 3.699300 -10.15707 180.8054 1.298628
## Test set      1.9753453993 30.03612 4.937974 339.99430 841.9069 1.733461
##                     ACF1
## Training set -0.0362758
## Test set             NA
```

Other models

We then built basic NNETAR, TBATS, and daily seasonal naive models to explore which can capture the trend and forecast best, compared to ARIMA models.

NNETAR

Neural network model (NNETAR) is know for capture non-linear trends and forecast price picks.

From the plot above (with number of total days in x-axis), we can see the models fails to forecast future price spikes.

```
##                       ME     RMSE       MAE          MPE       MAPE       MASE
## Training set  -0.1062637  15.3372  2.731614    -77.42721   138.0525  0.9589249
## Test set     -75.4918729 103.8230 81.231537 -21600.51080 22331.8411 28.5160911
##                    ACF1
## Training set 0.1880357
## Test set            NA
```

The auto model generates m = 24, p (nonseasonal lag) = 28, and P (seasonal lag) = 1, size = 14. This size (number of layers in this network) is unnecessarily high for a simple model, which can be further tuned if chosen. RMSE is performing relatively well compared to ARIMA.

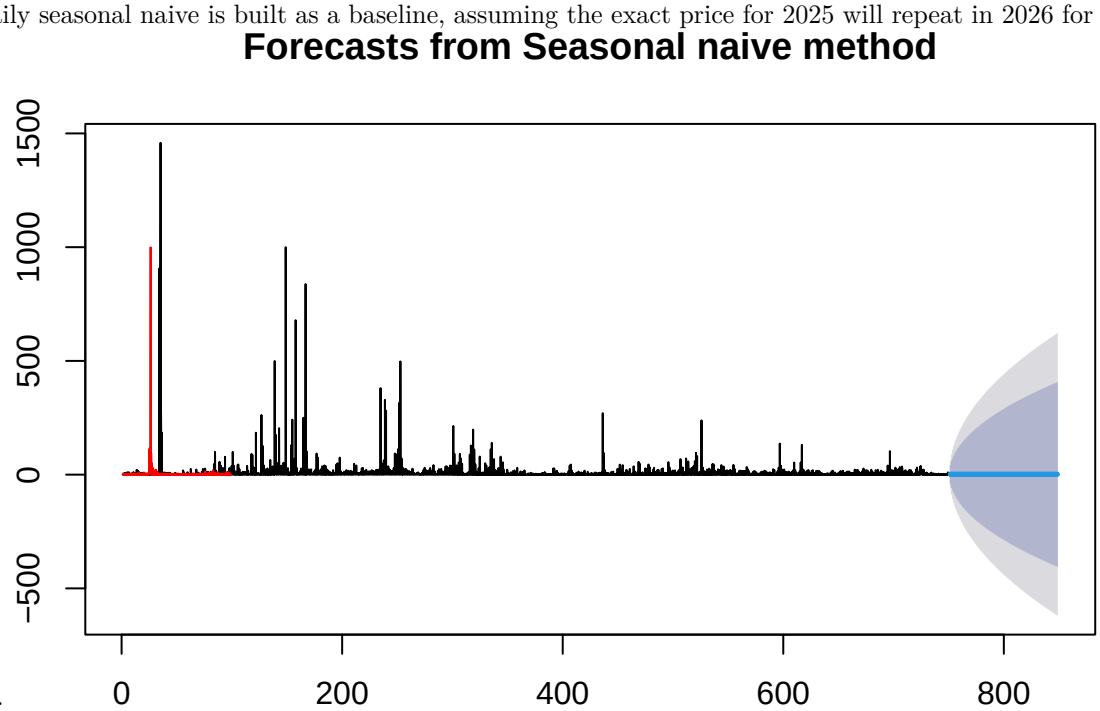
TBATS

TBATS may capture multi-seasonality, though smooth irregular patterns and noises (which is inherently an important part in RRS price); we explored how it may perform in forecasting for our goal. Lambda is set automatically.

```
##                     ME     RMSE      MAE       MPE     MAPE      MASE      ACF1
## Training set 0.5839835 17.26124 2.189309 -12.56823 36.36274 0.7685506 0.2349306
## Test set     2.3103995 29.96135 2.455188  12.55483 56.33663 0.8618866        NA
```

Seasonal Naive as a baseline comparison

Lastly, daily seasonal naive is built as a baseline, assuming the exact price for 2025 will repeat in 2026 for the same day.




```
##                        ME     RMSE      MAE       MPE      MAPE     MASE
## Training set -0.001356253 31.97748 3.596754 -28.93572  50.93494 1.262630
## Test set      1.736906888 29.85820 2.595176 -96.47556 120.26363 0.911029
##                    ACF1
```

```
## Training set 0.7073246
## Test set            NA
```

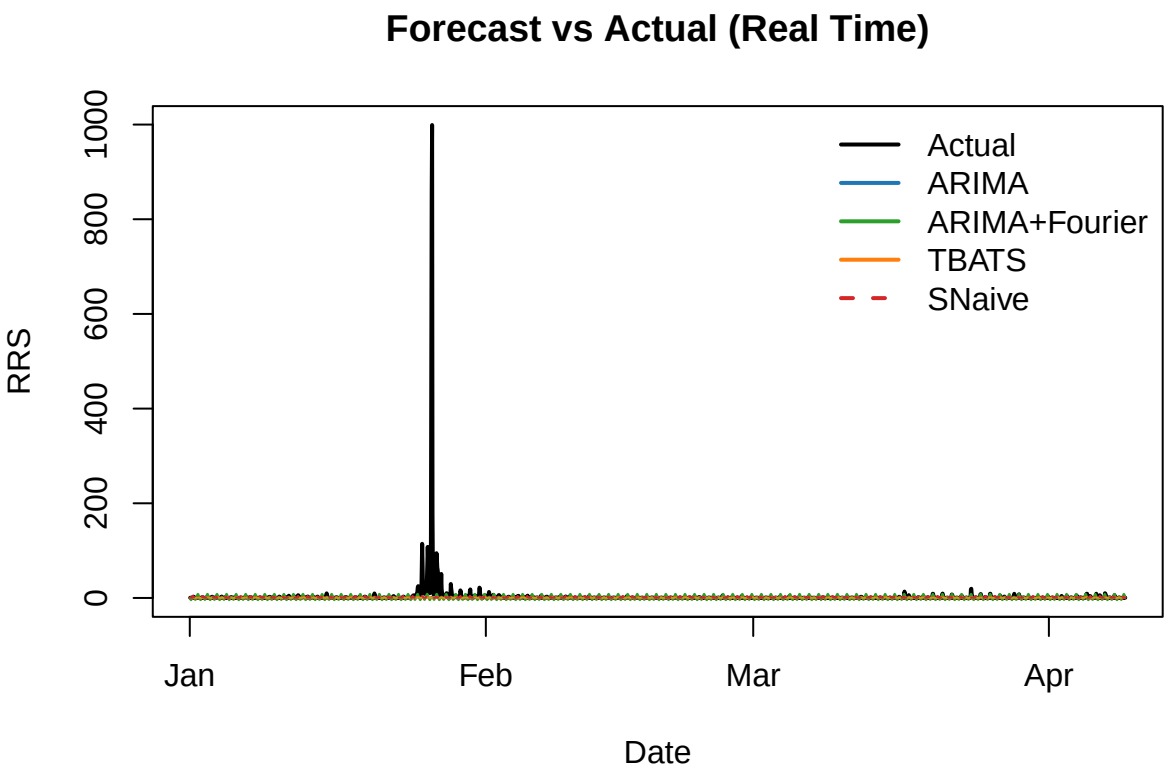
Unexpectedly, the seasonal naive model has a good RMSE on the test set compared to other more complex models, though higher RMSE for the training set. We will compare all models in later sections

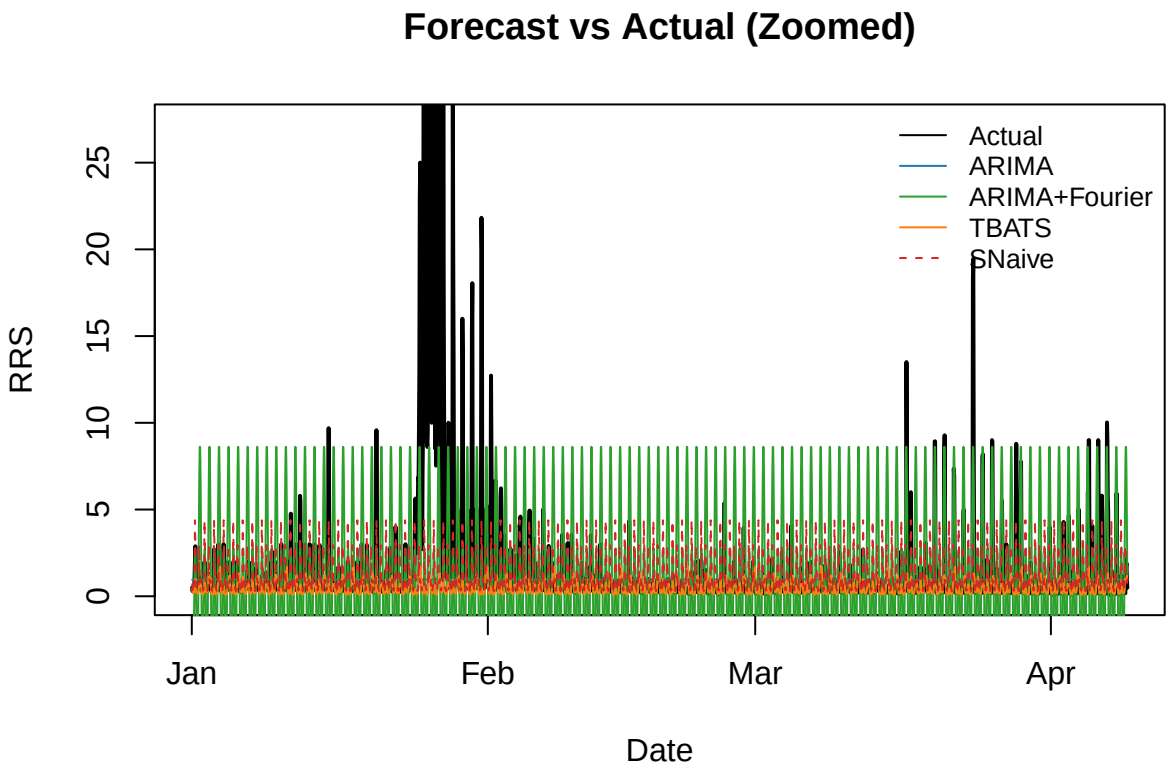
Result table

```
##                   RMSE    MAE   MASE
## ARIMA           29.941  2.671  0.938
## ARIMA_Fourier   30.036  4.938  1.733
## NNETAR         103.823 81.232 28.516
## TBATS           29.961  2.455  0.862
## SNaive_Daily    29.858  2.595  0.911
```

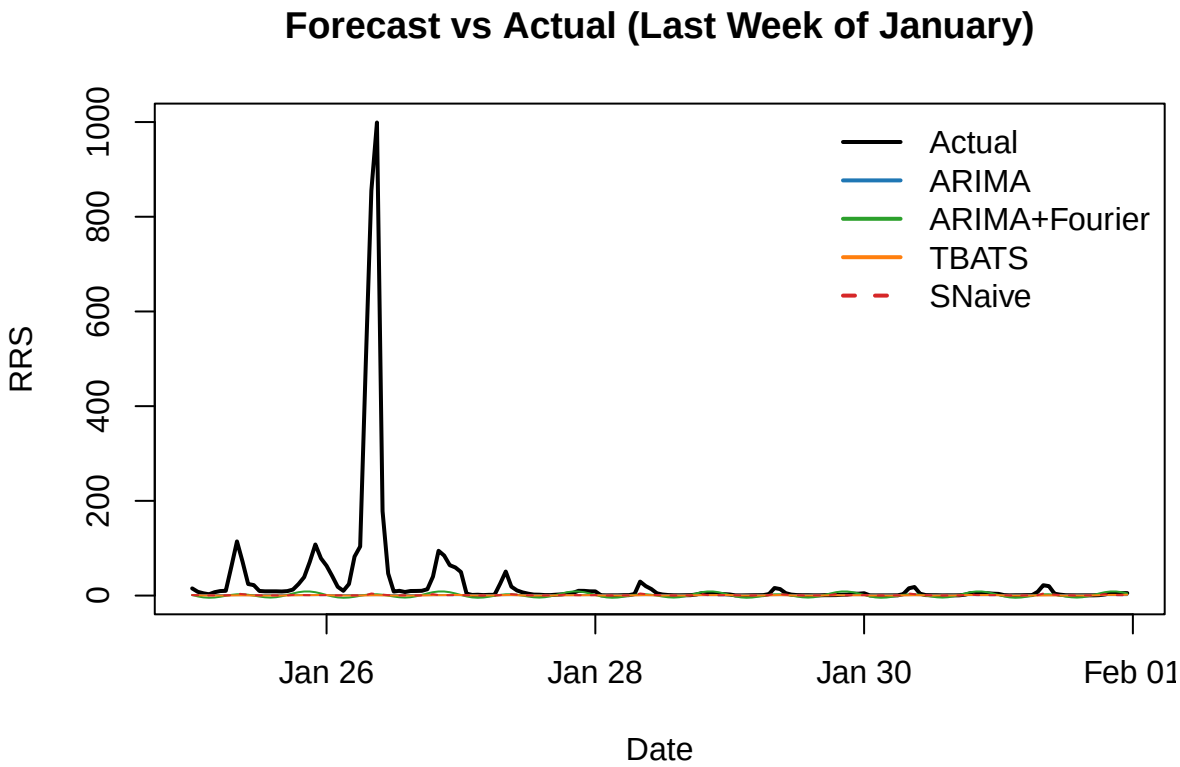
The results table indicates that TBATS performs best overall, achieving the lowest MAE and MASE by effectively capturing complex seasonality and fluctuations. The daily seasonal naïve model also performs well surprisingly, suggesting that RRS prices have a strong and consistent daily cycle. ARIMA provides reasonable accuracy, while NNETAR is moderate, and ARIMA with Fourier terms performs worst, likely due to its inability to capture sharp, irregular price movements typical in ancillary service markets.

Result plots



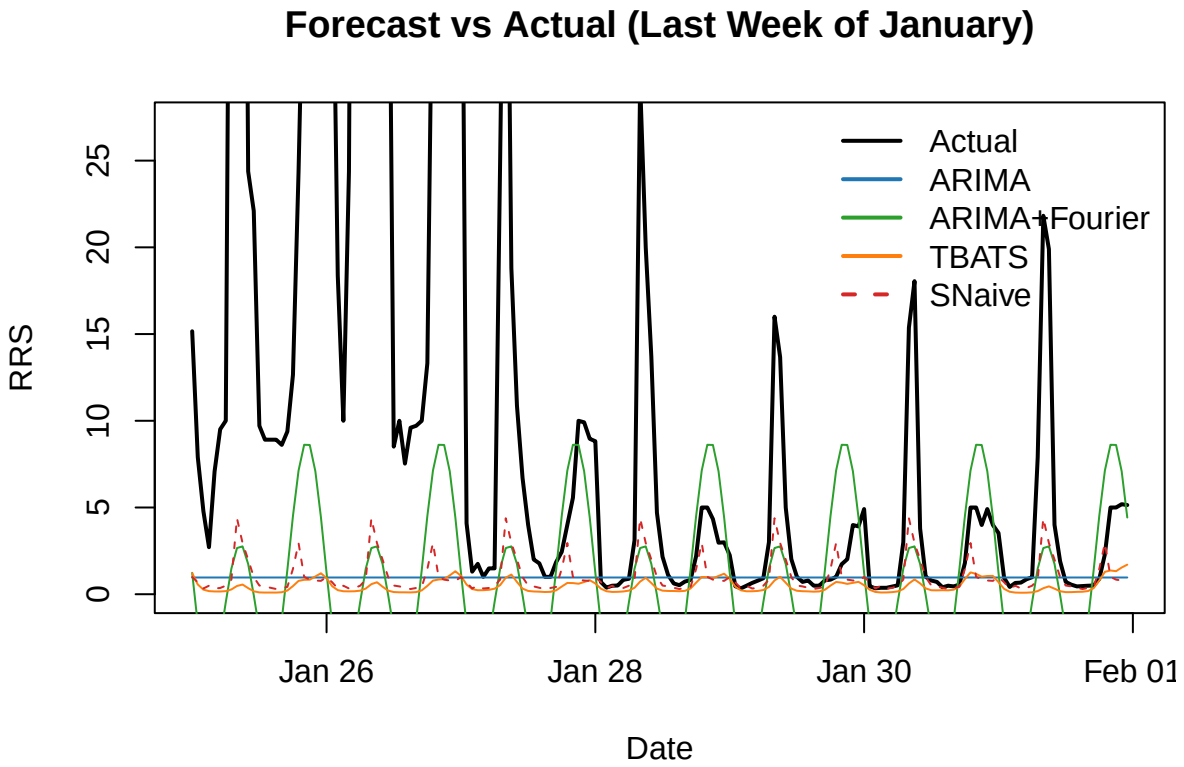


The two graphs above showed that over the full time horizon, RRS prices are generally low but exhibit a sharp spike in late January in 2026, reflecting a likely scarcity or system stress event in ERCOT. All models struggle to capture this extreme spike, with forecasts remaining relatively smooth and close to baseline levels. This highlights the difficulty of predicting rare, high-impact events in ancillary service markets, where price dynamics are largely stable but occasionally driven by sudden reliability needs.



Focusing on the last week of January reveals more detail around the volatility period. The actual series shows

rapid fluctuations and a pronounced spike, while the models track the general daily pattern but fail to match the magnitude of extreme movements. TBATS and ARIMA follow the underlying structure more closely, whereas the seasonal naïve model captures timing but underestimates variability, reinforcing that models can capture routine behavior but not sudden shocks.



In the zoomed-in view, the dominance of the spike becomes even clearer relative to typical price levels. Most forecasts remain tightly clustered near zero to moderate values, missing both the height and abruptness of the peak. This emphasizes that while models are effective at capturing regular seasonality, they systematically underestimate extreme price excursions, which are critical in RRS markets driven by rare but significant reliability events.

Summary and Conclusions

This study evaluates multiple time series models—ARIMA, ARIMA with Fourier terms, NNETAR, TBATS, and a seasonal naïve benchmark—for forecasting ERCOT's Responsive Reserve Service (RRS) prices. Results show that while all models achieve similar RMSE, TBATS performs best overall based on MAE and MASE, indicating stronger accuracy in capturing underlying patterns. The daily seasonal naïve model also performs competitively, suggesting that RRS prices exhibit strong and persistent daily seasonality. ARIMA provides stable, moderate performance, while NNETAR offers limited improvement, and ARIMA with Fourier terms performs worst, likely due to its inability to capture irregular dynamics. Visual analysis further reveals that all models struggle to predict extreme price spikes, which are characteristic of RRS markets.

The findings indicate that RRS prices in Texas are driven by a combination of stable daily patterns and occasional, high-magnitude volatility events. Models like TBATS, which can accommodate multiple seasonalities and flexible dynamics, are better suited for this environment, though even the best-performing models fail to anticipate rare spikes. From a practical perspective, simple models such as seasonal naïve can serve as strong baselines, while more advanced models provide incremental improvements in routine conditions but limited gains during stress events.

This study has several limitations. First, the models rely solely on historical price data and do not incorporate exogenous variables such as load, weather, or system conditions, which are key drivers of RRS price spikes.

Second, the evaluation focuses on aggregate error metrics, which may underrepresent model performance during extreme but economically significant events. Third, the sample size and time horizon may limit the ability to generalize results across different market regimes. Finally, all models considered are primarily statistical and may not fully capture the structural and regime-switching behavior inherent in ancillary service markets like ERCOT.